

NINE TAXONOMY DIMENSIONS

Mechanism-specific evaluation framework for AV/CAV congestion mitigation

1 Congestion mechanism

- Congestion onset / breakdown
- Capacity drop
- Queue formation
- Post-breakdown discharge
- Stop-and-go waves
- String instability
- Lane-changing friction
- Merging turbulence
- Physical spillback
- Shifted / redistributed congestion

2 AV/CAV decision lever

- Acceleration / deceleration
- Desired speed
- Desired headway / spacing
- Lane choice
- Lane-change timing / execution
- Merge / yield / gap acceptance
- Route choice
- Entry / access / release

3 Design method

- Rule-based / control law
- Optimal control
- Model predictive control (MPC)
- Game-theoretic control
- Reinforcement learning / MARL
- Heuristic control

6 Operating context

FACILITY CONTEXT

- Ring road / single-lane testbed
- Freeway; on-ramp merge; lane drop
- Weaving; work zone; sag
- Other geometry / facility-neutral

SPATIAL / SYSTEM SCALE

- Vehicle string / platoon
- Segment / bottleneck / facility
- Corridor / network / system

5 Operational application

- Traffic smoothing / mobile actuation
- Speed harmonization / dynamic headway
- Bottleneck-inflow regulation
- Lane assignment / lane-use control
- Cooperative merging / gap creation
- Ramp coordination
- Integrated longitudinal-lateral control
- Platooning
- Routing / DTA / managed-lane operation
- Perimeter / corridor control; fleet rebalancing

4 Coordination/implementation mode

- Individual/onboard
- Cooperative
- Infrastructure-assisted
- Centralized
- Distributed/decentralized
- Hybrid vehicle-infrastructure

7 Mixed-traffic and deployment conditions

COMPOSITION / CONTROL AUTHORITY

- AV/CAV penetration; controllable share
- Connected HDVs; heavy vehicles

HUMAN RESPONSE / CONTROLLERS

- HDV following; lane change / cut-in
- Controller heterogeneity; compliance

COMMUNICATION / DEPLOYMENT

- Reliability / latency; sensing / state
- Fallback / ODD limits

8 Evidence setting

- Analytical / theoretical
- Simulation
- Test-track experiment
- Field experiment
- Observational / open-road data
- Network / demand model

9 Congestion metric

- Onset time; breakdown probability
- Sustainable pre-breakdown flow
- Capacity; post-breakdown discharge
- Bottleneck outflow; throughput
- Capacity drop; recovery time
- Queue length / discharge; speed variance
- Wave amplitude; delay; travel time
- Storage / blocking / spillback
- Network flows; VMT / VKT; system travel time
- Delay distribution; safety surrogate measures

LOCAL-TO-NETWORK FEEDBACK

9 Observed outcomes

Flow; discharge; queues;
delay; travel cost

Behavioral response

Route; lane; departure time;
fleet movement; demand

Changed network state

Bottleneck arrivals; composition;
downstream queues; spillback

5-6 Reassess application and operating context

Persistent benefit, physical spillback,
or shifted / redistributed congestion?